



WPI

A COMBINED EXPERIMENTAL AND COMPUTATIONAL STUDY OF THE CELL WALL PROPERTIES IN TIP-GROWING *PHYSCHOMITRIUM PATENS*



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Abstract

An uptake in the interest of tip-growing cells has occurred, due to their display of unique cell wall mechanics. Specifically, the model organism *P. patens* is characterized by a prevailing attribute of polarized growth during the protonemal stage, a process in which the cell elongates in a self-similar manner. Due to the nature of the turgor pressure, it has been theorized that an elastic gradient pertains towards the tip of the cell, with previous research supporting said characteristic in fission yeast. In addition, it has been theorized that the tip of the organism has an acidic pH while in this growth phase. As seen in cucumber cells, expansin proteins that utilize microfibril sliding to loosen the cell wall require a low-pH environment for activation, which is caused by proton pumps secreting positively charged hydrogen ions. Both properties were investigated using confocal imaging of developing protonemata of *P. patens*. Stimulated plasmolysis enabled triangulation of surface gradients to identify elastic identity, while overall shrinking conformation confirmed the efficacy of the addition of mannitol solution. Spinning disk microscopy imaging pointed to the cell wall fluorescing at a higher capacity along the tip of the cell; yet, distinct controls led to further investigation. However, we did not observe the expected flat fluorescence profile in the control cases. Overall, our research leads to further discussion on the nature of the cell wall during suspension and growth.

Background

1. Plants and the Cell Wall

a. Importance

The Cell Wall is Highly Flexible and Adaptable

Resistant to Bursting

Constantly Able to Reform

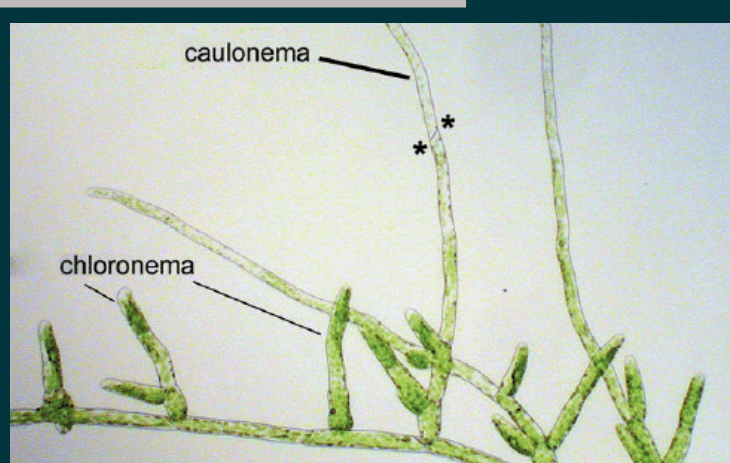
Largest Source of Renewable Carbohydrates on Earth

The ability to understand the mechanical properties aids in the knowledge, extraction and modification of the wall

Cellulosic Biofuels

Dessication Tolerance

BioMass

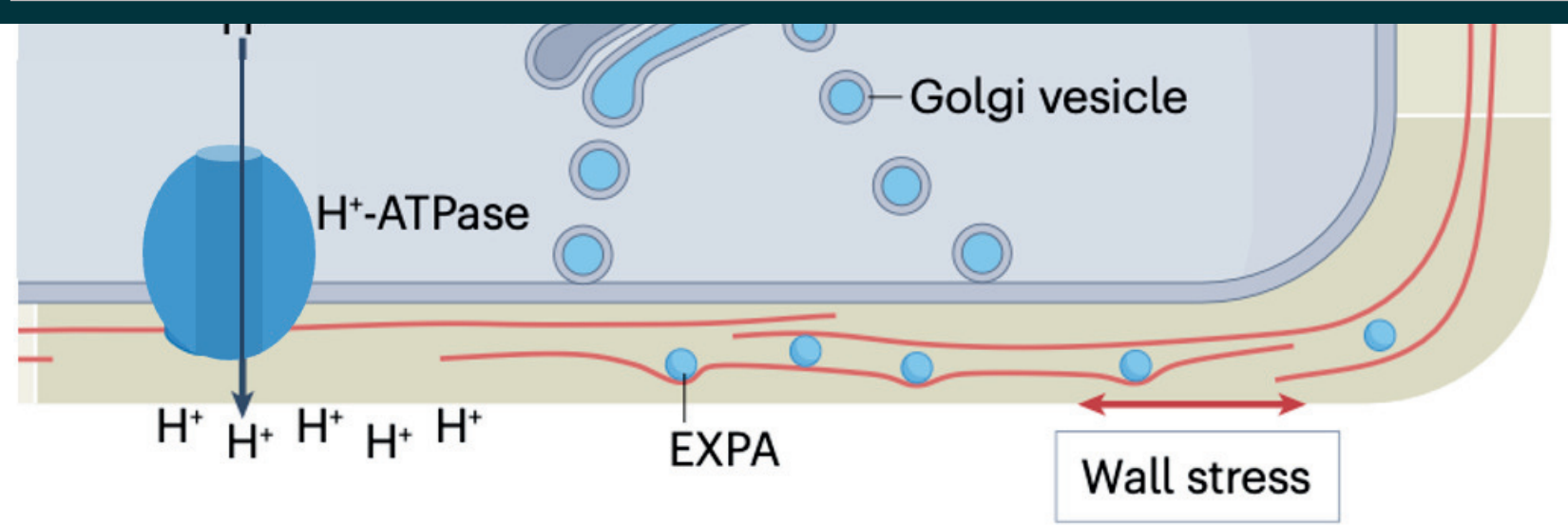


Caulonema

- Smaller width
- Spread out chloroplasts
- Nutrition acquisition & localization

Chloronema

- Larger width
- Bunched chloroplasts
- Photosynthesis & spore germination

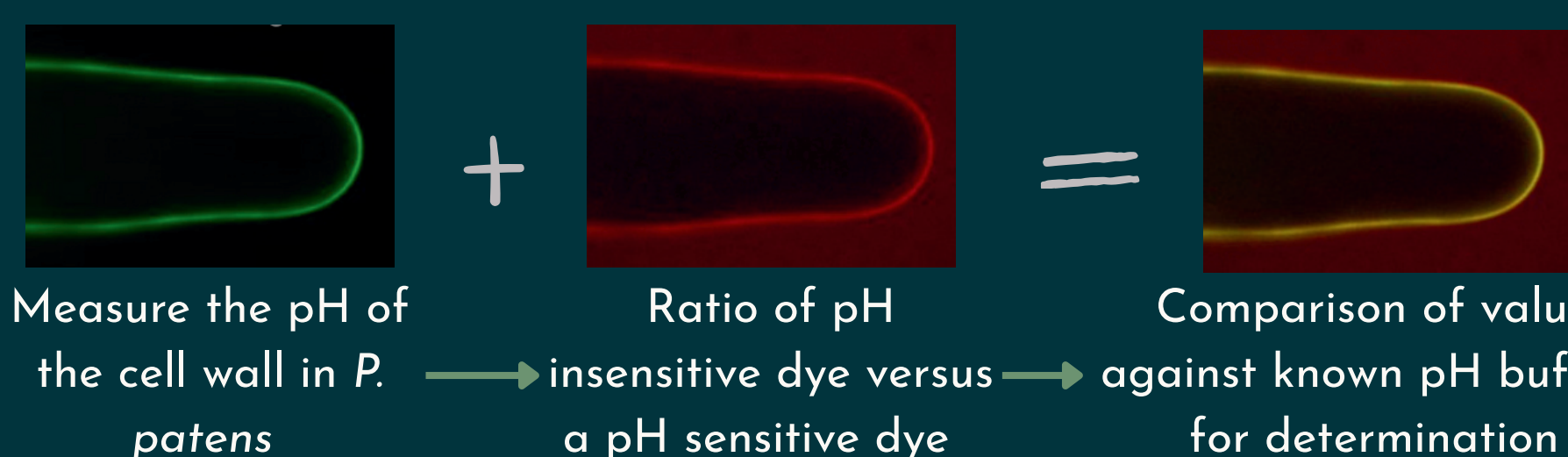


1. Proton pumps excrete hydrogen ions

2. Low pH activates expansin (EXPA) proteins

3. As a result, microfibril sliding occurs

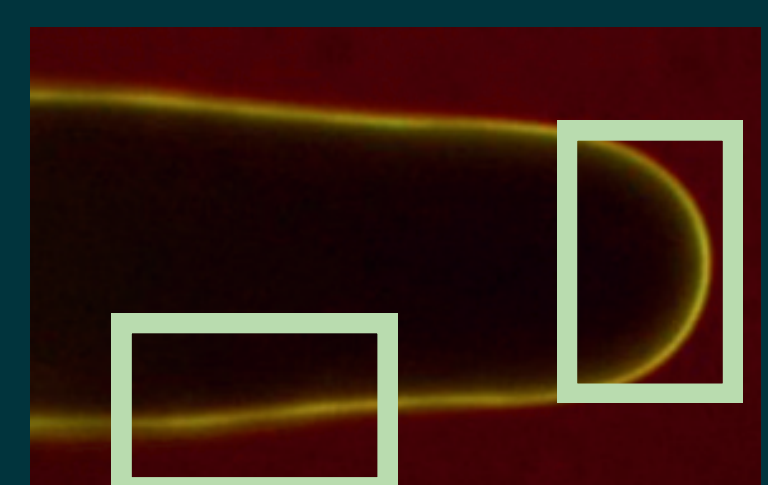
b. Confirmation of Expansin Activation



Hypotheses

1. Wall Mechanics

We predicted the elasticity of *P. patens* cell wall contains a gradient. This arises from a difference in wall material arrangement at the tip

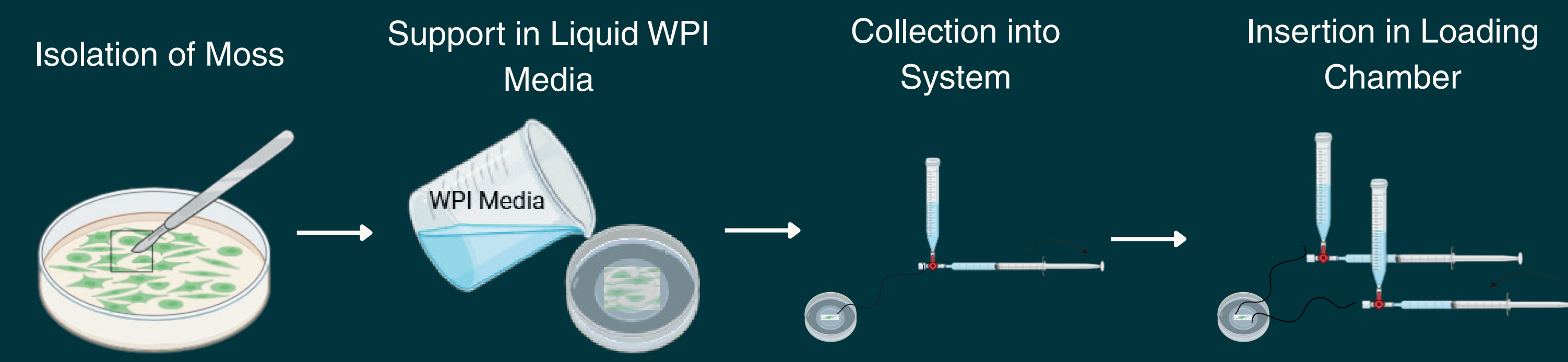


2. pH Experiment

We predict that there is a higher amount of expansin activity at the tip of the growing cell which will result in a looser wall material

Schematics for Preparing and Analyzing *P. patens* Cell

A. Loading of Protonemata



B. Moss Labeling Methods

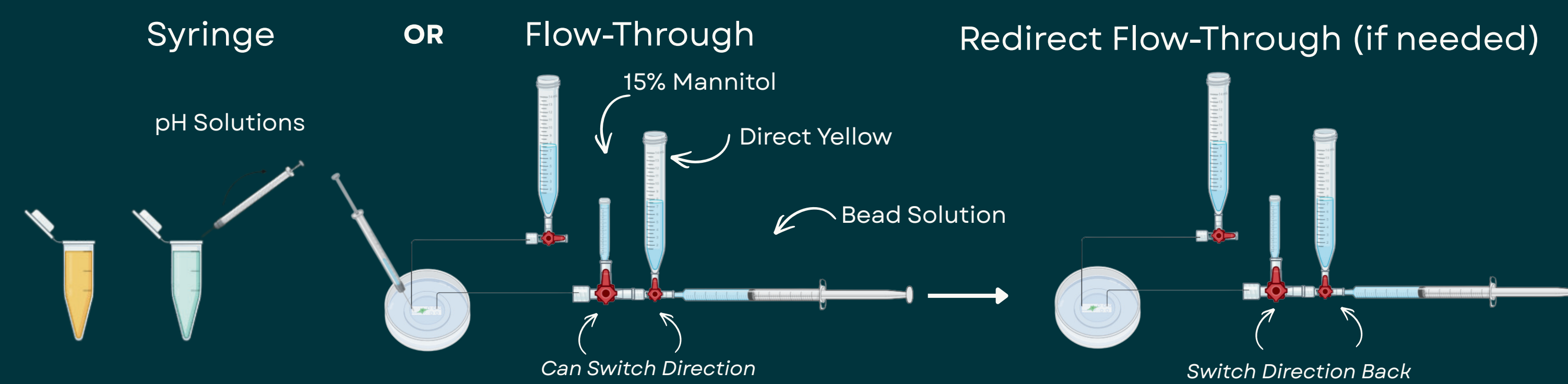


Figure 1 Schematic of Plant Loading into Microfluidic Devices: A. Moss was obtained from a cultivated *P. patens* plate. Isolated protonemata cells were submerged in liquid WPI media. A selected protonemata cell cluster was pulled up using a syringe. The plant was loaded into the device. The device was set up under a microfluidic flow-through. B. pH solutions were collected in a syringe and injected into the loading chamber. The microfluidic device was reattached to the flow-through mechanism, or the flow-through mechanism was switched to allow the desired solution through, then readjusted to its original state if required.

Plasmolysis Reveals Underlying Cell Wall Elasticity Through Shrinkage

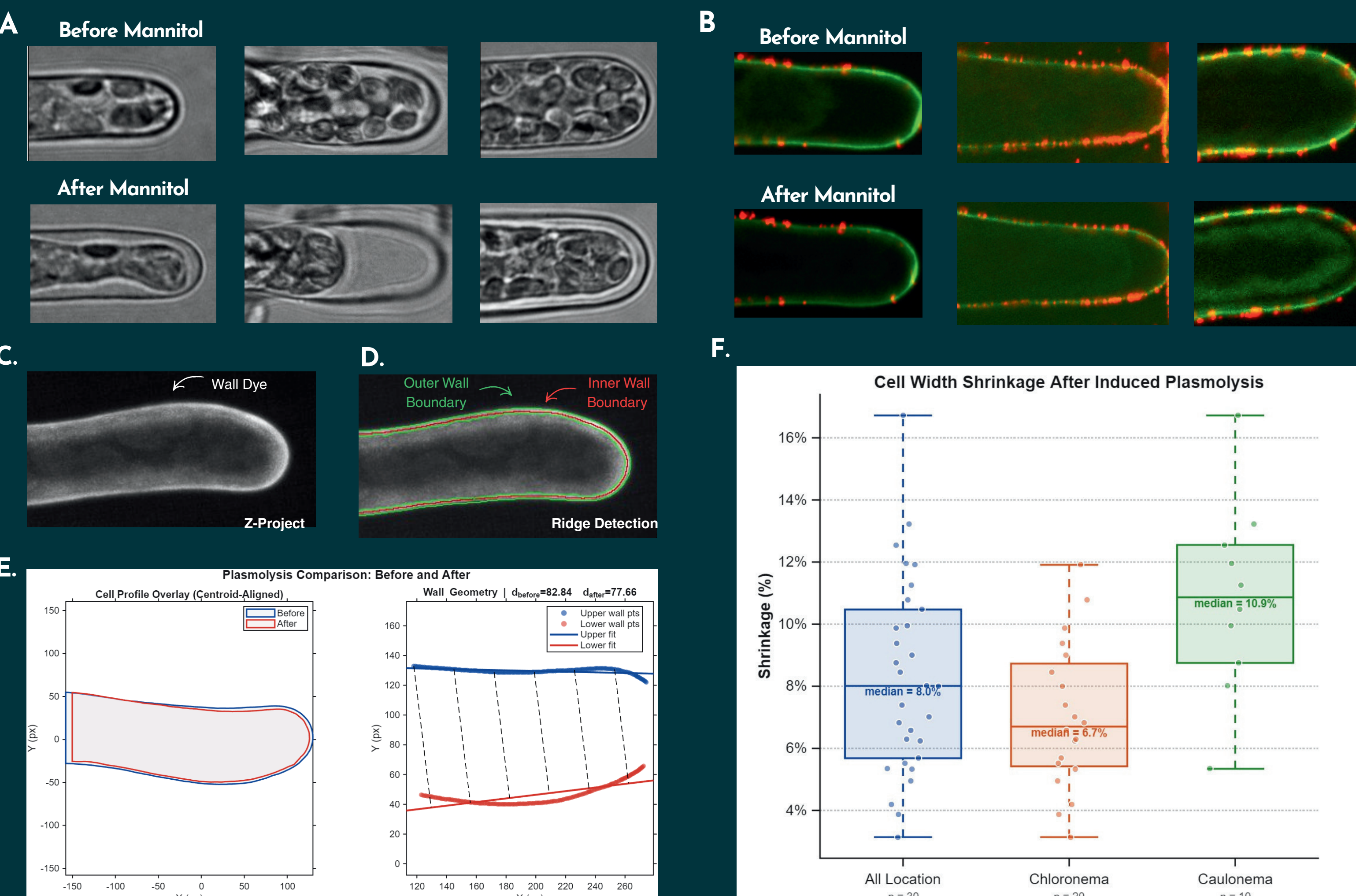


Figure 2 Quantification process of cell wall width after induced plasmolysis: (A) Brightfield imaging of cell shrinkage of *P. patens* cells before and after addition of filtered 15% mannitol with WPI media. (B) Imaging of fluorescent beads and cell wall labeling joint under 488nm (green) and 561nm (red) emission wavelengths (C) Z-project of a caulonema cell labeled with Direct Yellow wall dye. (D) Ridge detection outlines overlaid on a Z-projected image of a caulonema cell. Green and red lines denote the outer and inner wall boundaries respectively. (E) Quantitative analysis of cell wall shrinkage following plasmolysis. (Left) Centroid-aligned overlays of the inner wall boundary coordinates before (blue) and after (red) plasmolysis treatment (Right) Linear regression fits to the upper and lower wall segments used for wall thickness estimation. (F) Quantification of shrinkage percentages of all locations (n=30), caulonema (n=10) and chloronema (n=20) locations undergoing plasmolysis.

Discussion + Conclusion

- Mannitol addition leads to a shrinkage of the *P. patens* cell wall, with caulonema cell walls shrinking at a higher average percentage than chloronema cells
- While the results of the pH are still preliminary, the fluorescence gradient suggests that the tip of *P. patens* cell has a more acidic pH than the edges of the wall
- Fluorescent dyes did not flatten as well as expected, needing additional testing

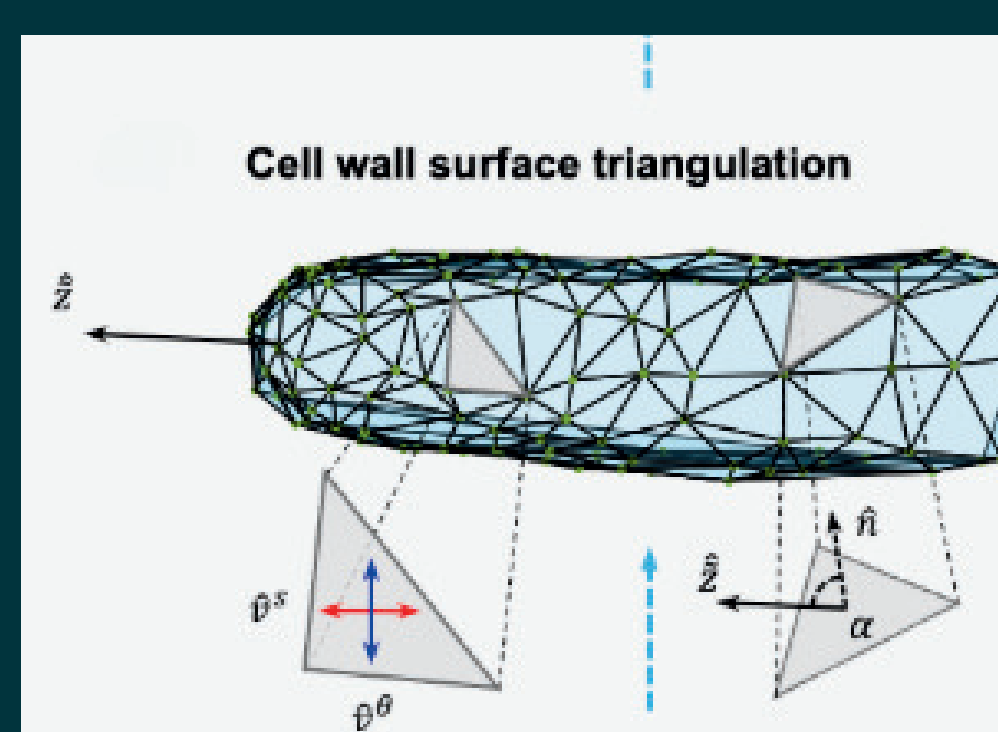


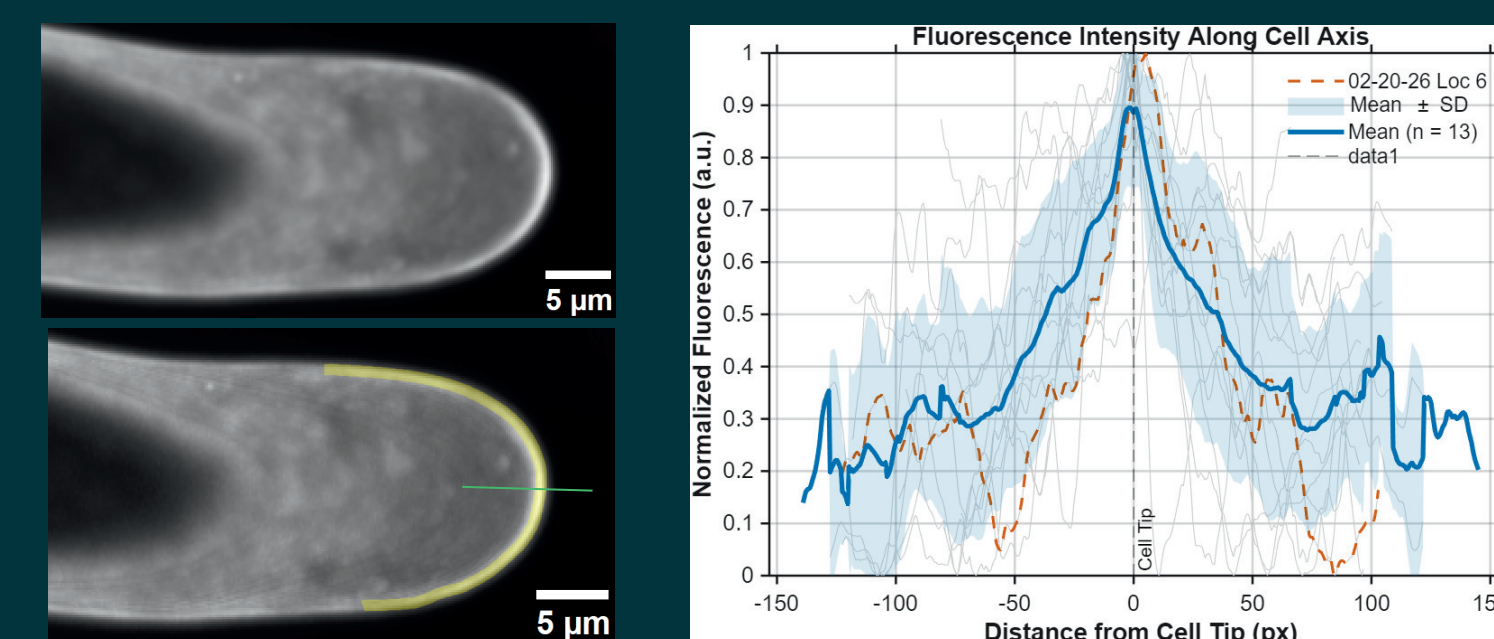
Figure 9. Experimental Modeling of the Cell Wall in 3D Cell Wall Surface triangulation mapping of fluorescent beads onto the 3D Surface Model

- Experiments with bead tracking can be used for a cell wall surface triangulation method
- The work above contributed to the methodology that is now published

Distinct Fluorescent Labeling Approaches on *P. patens* Cell Wall

a. Baseline

pH Dependent Dye: pHrodo Green



- pHrodo Green is a pH sensitive dye which peaks at a low pH
- Fluorescence identity shows peak fluorescence at the tip region, seen in both the mean and the individual example
- A fluorescence gradient is seen from the tip to the side until a steady fluorescence occurs once the side is reached

Figure 4: pHrodo Green Labeling of *P. patens* protonemata (Left) Max project of spinning disk confocal z-stack in 488nm channel and line segment for average pixel value. (Right) Mean graph of n=13 locations of *P. patens* labeled with pHrodo green.

b. Controls

#1: pH Independent Dye: Alexa Fluor

- Alexa Fluor is a pH-independent dye
- The fluorescent profile shows that the side and tip of the cell wall are nearly equivalent
- Regions between the tip and the side of the cell decrease

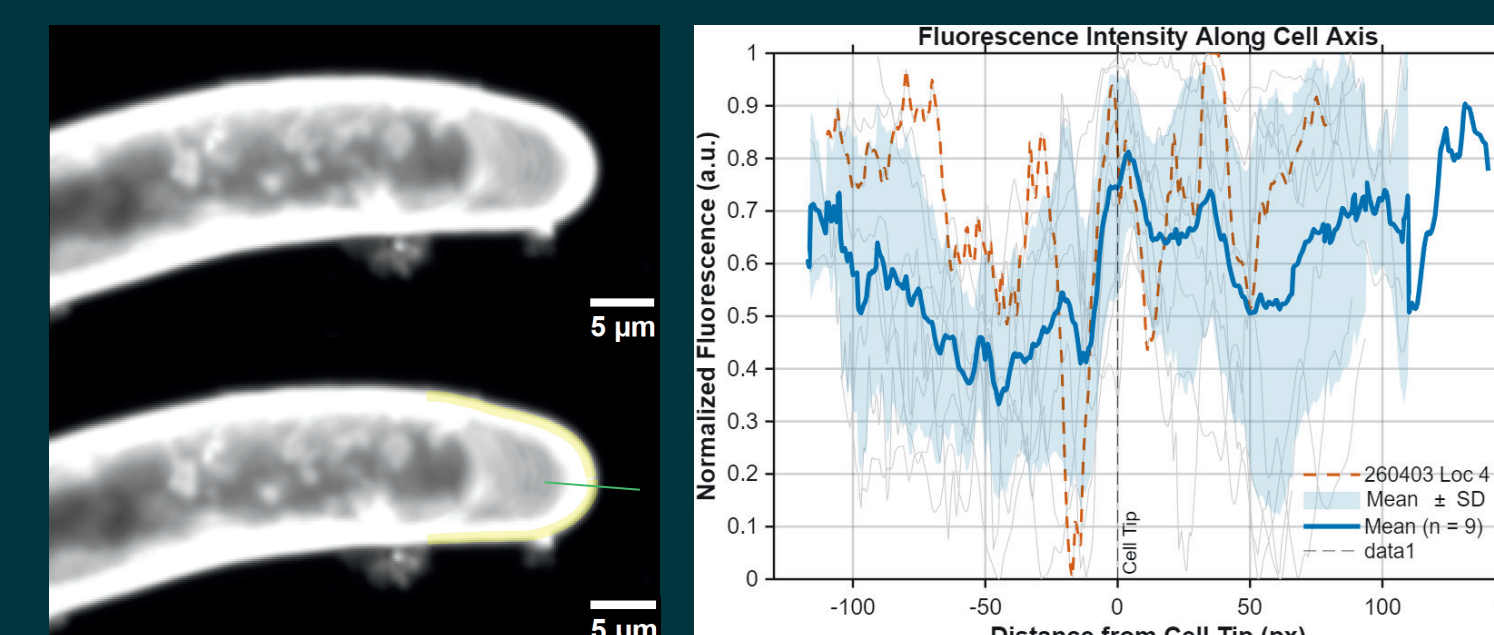


Figure 5: Alexa Fluor Labeling of *P. patens* protonemata (Left) Max project of spinning disk confocal z-stack in 488nm channel and line segment for average pixel value. (Right) Mean graph of n=9 locations of *P. patens* labeled with Alexa Fluor.

#2: Mannitol Addition Pre pHrodo Green Dyeing

- Mannitol induces the removal of turgor pressure
- Leads protonemata into a state of halted growth
- Profile flattens slightly compared to pHrodo Green from above; however, not a full removal of the pH gradient

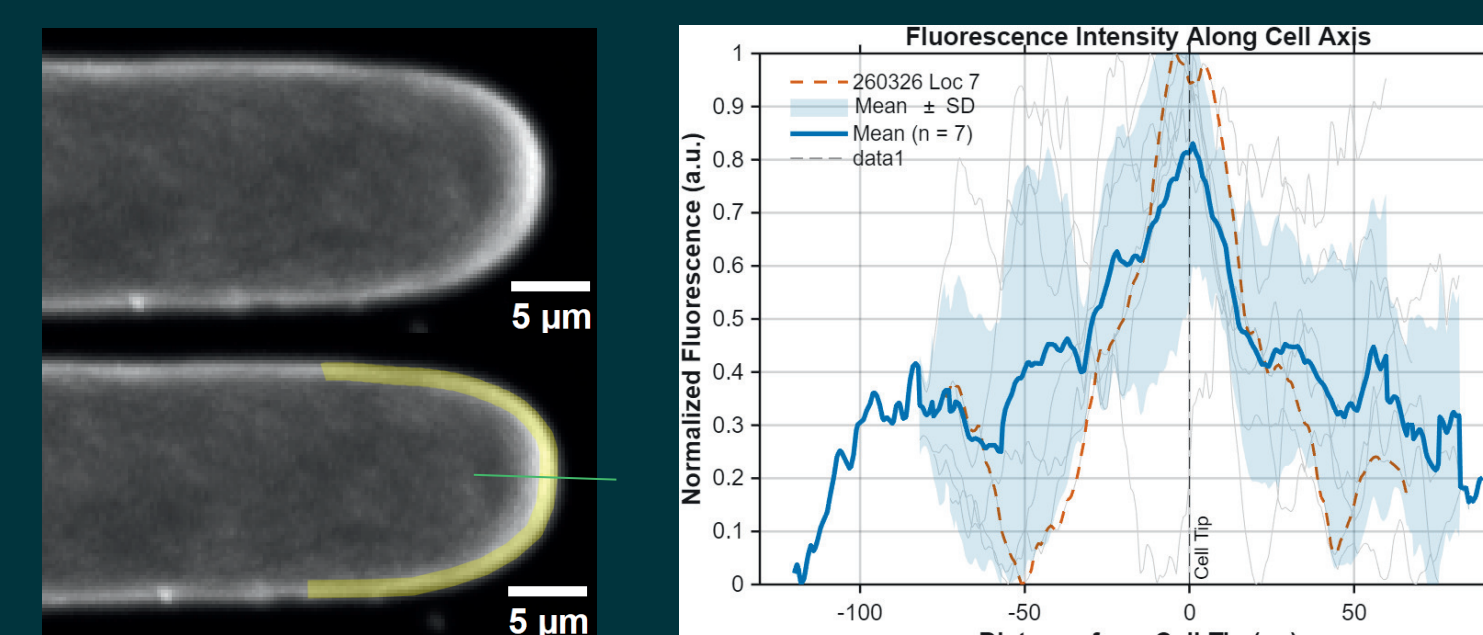


Figure 6: Labeling of *P. patens* protonemata post Mannitol induction (Left) Max project of spinning disk confocal z-stack in 488nm channel and line segment for average pixel value. (Right) Mean graph of n=7 locations of *P. patens* labeled with pHrodo Green after Mannitol induction

#3: pHrodo Green Labeling Before Mannitol

- pHrodo was added before, and then re-added once again after the flow-through of Mannitol
- Profile flattens slightly compared to pHrodo Green from above; however, a removal of the pH gradient was not achieved

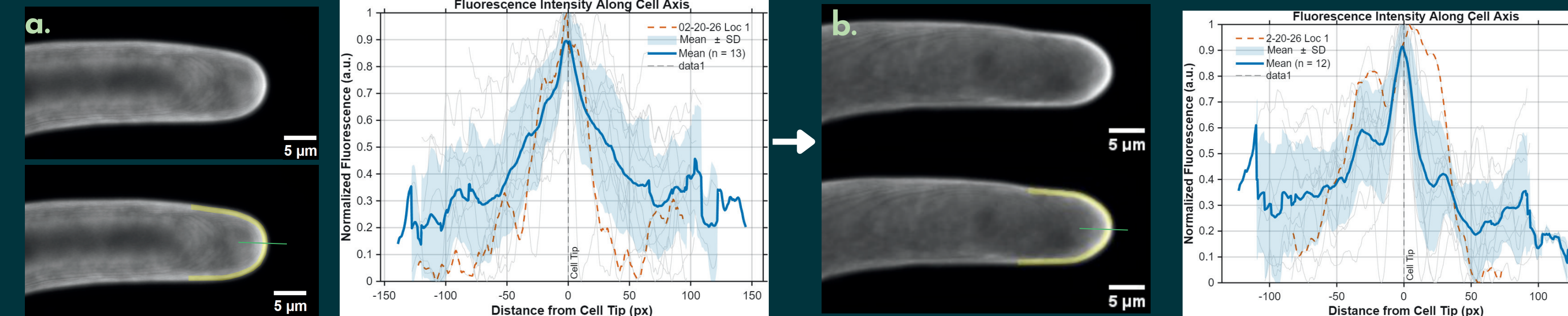


Figure 7: pHrodo Green Labeling of *P. patens* protonemata before and after mannitol treatment (a) Before mannitol (b) After mannitol. (Left) Max project of spinning disk confocal z-stack in 488nm channel and Line segment for average pixel value. (Right) Mean graph of n=13 locations of *P. patens* labeled with pHrodo Green before Fusicoccin addition versus after Fusicoccin addition (n=12).

#4: pHrodo Green Labeling Before Fusicoccin

- Fusicoccin activates the proton pumps all around the cell, leading to a lower pH around the entirety of the cell wall
- Such an action, in turn, is hypothesized to lead to a removal of the pH gradient
- pHrodo Green was added before, and then the cell was treated with Fusicoccin, leading the cell wall to swell
- Profile flattens at the tip towards the near edge of the cell; however, it drops at the side of the cell wall

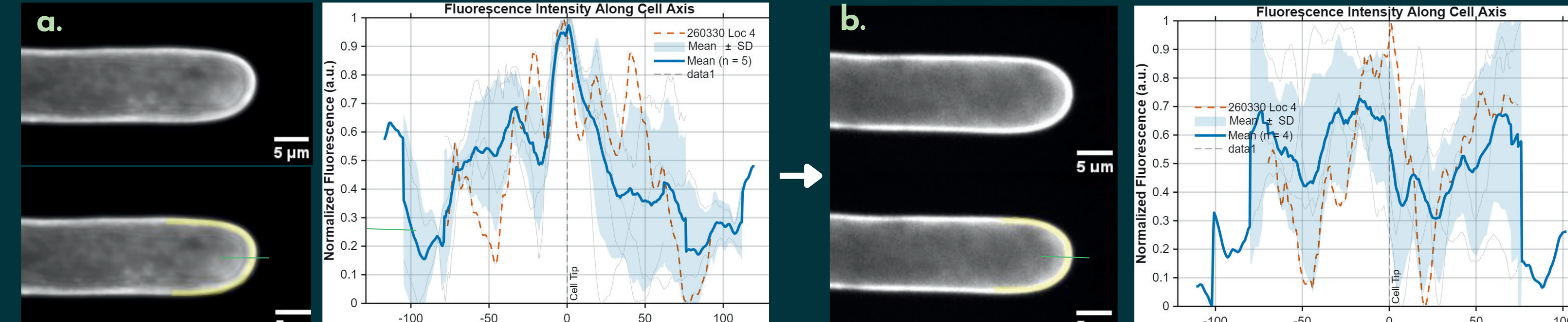


Figure 8: pHrodo Green Labeling of *P. patens* protonemata before and after fusicoccin treatment (a) Before Fusicoccin (b) After fusicoccin. (Left) Max project of spinning disk confocal z-stack in 488nm channel and line segment for average pixel value. (Right) Mean graph of n=5 locations of *P. patens* labeled with pHrodo Green before fusicoccin addition versus after fusicoccin addition (n=4).

Future Work

- Acquisition of more fluorescent data values for each control to validate current findings
- Calculation of pHrodo Green to Alexa Fluor ratio for pH identity
- Statistical conformation of the fluorescence distributions
- pH distinction between caulonema versus chloronema cells

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